

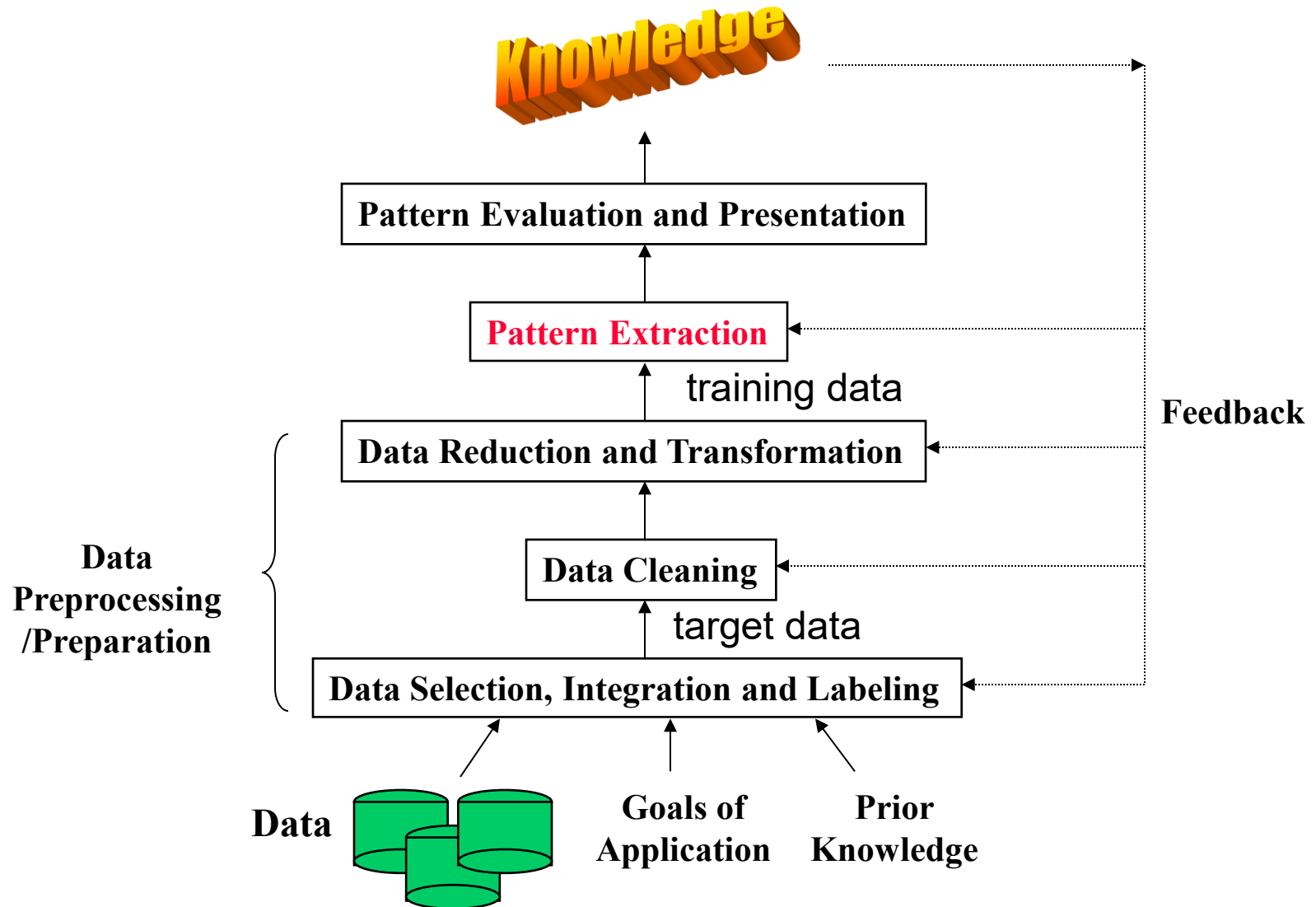
Data Mining

Data Preprocessing/Preparation

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Process of Data Mining and KDD



Outline

- ▶ Why preprocess the data?
- ▶ Data integration
- ▶ Data cleaning
- ▶ Data reduction
- ▶ Data augmentation
- ▶ Data labeling
- ▶ Data transformation
- ▶ Discretization

Why Data Preprocessing?

- ▶ Heterogeneous data – data integration
 - ▶ From various departments, in various forms
- ▶ Data not labeled – Data Labeling
 - ▶ None or a small portion of examples have true outputs.
- ▶ Dirty data – data cleaning
 - ▶ Incomplete data: missing attribute values
 - ▶ e.g., occupation=""
 - ▶ Noisy data: containing errors
 - ▶ e.g., Salary="-10"
 - ▶ Discrepancies in codes or names
 - ▶ e.g., US=USA

Why Data Preprocessing? (Cont'd)

- ▶ Data not in the right format – data transformation
 - ▶ Normalization, discretization, etc.
- ▶ A huge amount of data – data reduction
 - ▶ Speed up mining
- ▶ Data set is too small or not diverse – data augmentation
 - ▶ To learn a better model

No quality data, no quality mining results!

Major Tasks in Data Preprocessing

- ▶ Data integration
 - ▶ Integration of multiple databases or files
- ▶ Data labeling
 - ▶ Provide true outputs (labels) for training examples
- ▶ Data cleaning
 - ▶ Fill in missing values, identify outliers and smooth out noisy data, and resolve discrepancies
- ▶ Data transformation
 - ▶ Feed the right data to the mining algorithm
- ▶ Data reduction and augmentation
 - ▶ Obtains reduced representation in volume but produces the same or similar analytical results, or
 - ▶ Creates more examples if the data set is too small
- ▶ Data discretization
 - ▶ Part of data reduction and data transformation but with particular importance, transform numerical data into symbolic (discrete) data

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Data Integration

- ▶ Data integration:
 - ▶ combines data from multiple sources into a coherent store
- ▶ Schema integration
 - ▶ integrate metadata from different sources
 - ▶ *Entity identification problem*: identify real world entities from multiple data sources, e.g., $A.cust-id \equiv B.cust-\#$
- ▶ Detecting and resolving **data value conflicts**
 - ▶ for the same real world entity, attribute values from different sources are different
 - ▶ possible reasons: different representations, different scales,
 - ▶ e.g., hotel price in different currencies, metric vs. British units
 - ▶ e.g., Age="42", Birthday="03/07/1997"
 - ▶ e.g., Was rating "1,2,3", now rating "A, B, C"

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Data Cleaning

▶ Why is data dirty?

▶ Incomplete data come from

- ▶ human/hardware/software problems (e.g. equipment malfunction)
- ▶ different consideration between the time when the data was collected and when it is analyzed.
 - ▶ certain data may not be considered important at the time of entry

▶ Noisy data come from the process of

- ▶ data collection
- ▶ data entry
- ▶ data transmission

▶ Data cleaning tasks

- ▶ Fill in missing values
- ▶ Identify outliers and smooth out noisy data

How to Handle Missing Values in Pre-processing?

- ▶ Fill in the missing value manually: tedious + infeasible?
- ▶ Ignore the tuple containing missing values:

Cust-id	Age	Gender	Income	Credit
1	36	M	\$54K	good
2	24	M	\$20K	bad
3	37	M	\$50K	?
4	23	F	\$30K	good
5	55	F	\$25K	good
6	35	?	\$16K	bad
7	33	F	\$10K	bad

- ▶ usually done when class label is missing (assuming the task is classification)
- ▶ not effective when missing values in attributes spread in many different tuples.
- ▶ Fill it in with a value “unknown”
 - ▶ patterns containing “unknown” is ugly

How to Handle Missing Values in Pre-processing ? (*Contd.*)

- ▶ Global estimation
 - ▶ the attribute mean/median for numeric attributes
 - ▶ the most probable value for symbolic (i.e. categorical) attributes
- ▶ Local estimation: smarter
 - ▶ the attribute mean/median for all the tuples belonging to the same class (for numeric attributes)
 - ▶ the most probable value within the same class (for symbolic attributes)

Cust-id	Age	Gender	Income	Credit
1	36	F	\$55K	good
2	24	?	\$20K	bad
3	37	F	\$50K	good
4	23	F	\$30K	good
5	55	F	\$25K	good
6	35	M	?	bad
7	33	M	\$10K	bad

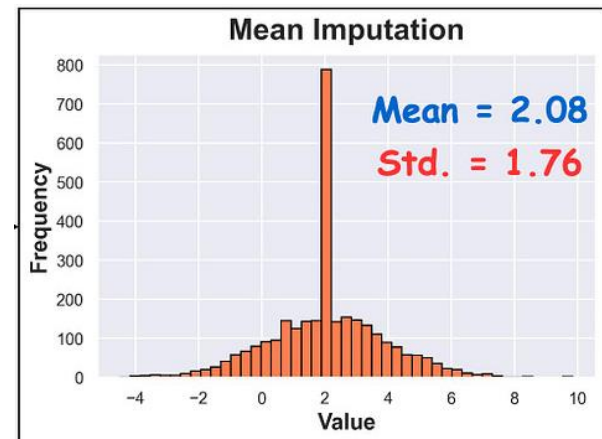
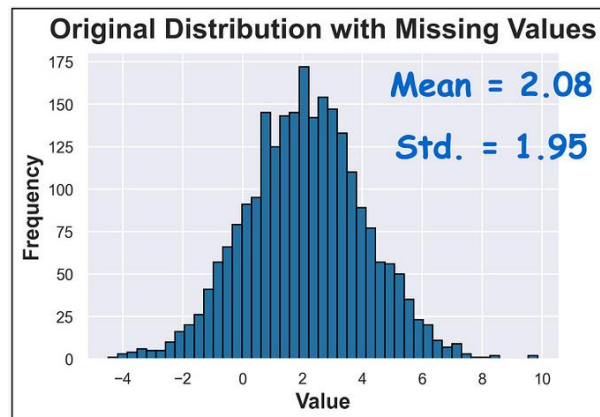
Disadvantage:

Replacing (imputing) missing values with mean or any other fixed value:

- alters summary statistics
- changes the distribution
- inflates the presence of a specific value

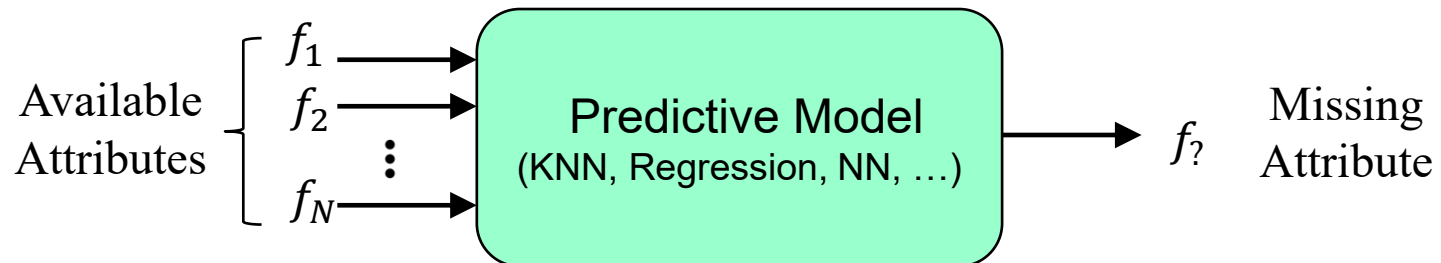
This can lead to:

- inaccurate modeling
- incorrect conclusions



How to Handle Missing Values in Pre-processing ? (*Contd.*)

- ▶ Use inference-based prediction techniques, such as
 - ▶ Nearest-neighbor estimator, decision tree, regression, neural network, etc.
 - ▶ good method with overhead



- ▶ A predictive model like a decision tree or KNN can estimate missing feature values by learning from other available features and predicting the most likely value for the missing one.

Noisy Data

- ▶ Noise: random error or variance in a measured variable
- ▶ Incorrect attribute values may be due to
 - ▶ faulty data collection instruments
 - ▶ data entry problems
 - ▶ data transmission problems

How to Handle Noisy Data?

▶ Clustering

- ▶ detect and remove outliers (An outlier is a value that does not follow the general pattern of the rest)

▶ Regression

- ▶ smooth by fitting the data into regression functions

▶ Using Z-score

▶ Moving average

- ▶ Use the arithmetic mean of neighborhood examples

▶ Data generalization or discretization:

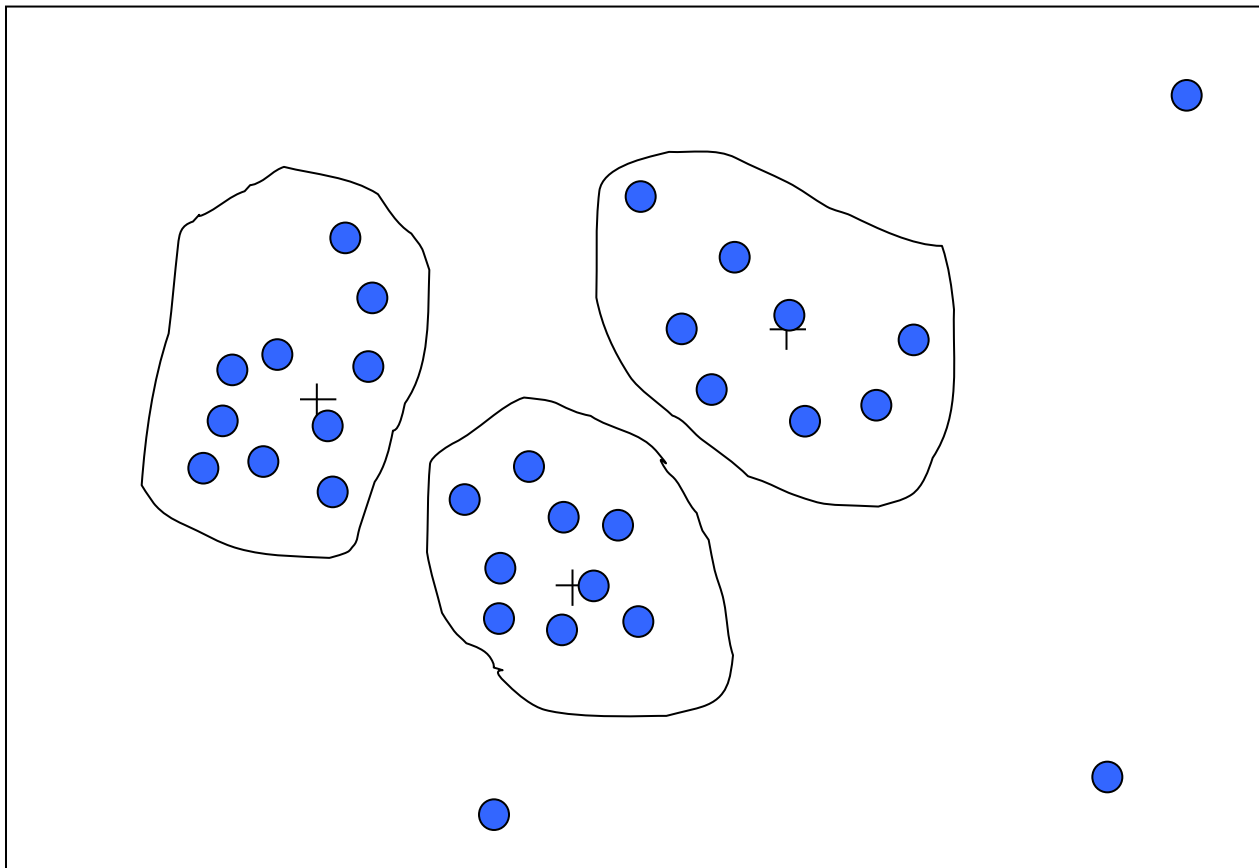
- ▶ first sort data and partition it into bins
- ▶ then one can smooth by bin labels, etc.

▶ Combined computer and human inspection

- ▶ detect suspicious values and check by human (e.g., deal with possible outliers)

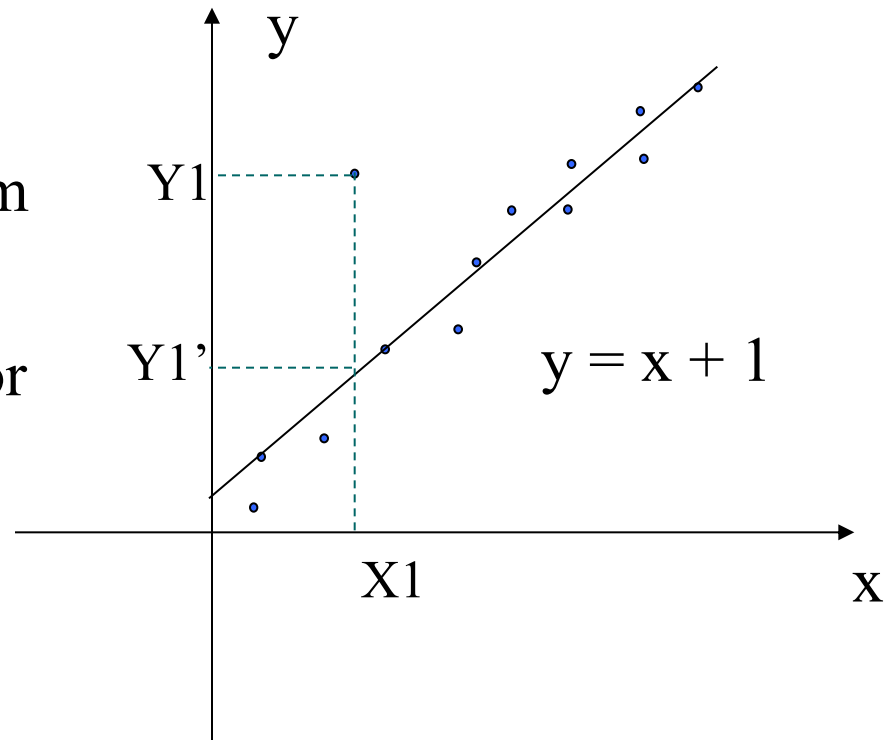
Cluster Analysis

- Clustering detects outliers as points far from any cluster center, which can then be removed or corrected to handle noisy data.



Regression

- ▶ Fit the data to a function.
- ▶ Data points too far away from the function are outliers.
- ▶ A *single linear regression*, for instance, finds the line to fit data with 2 variables so that one variable can predict the other.
- ▶ More variables can be involved in *multiple linear regression*.

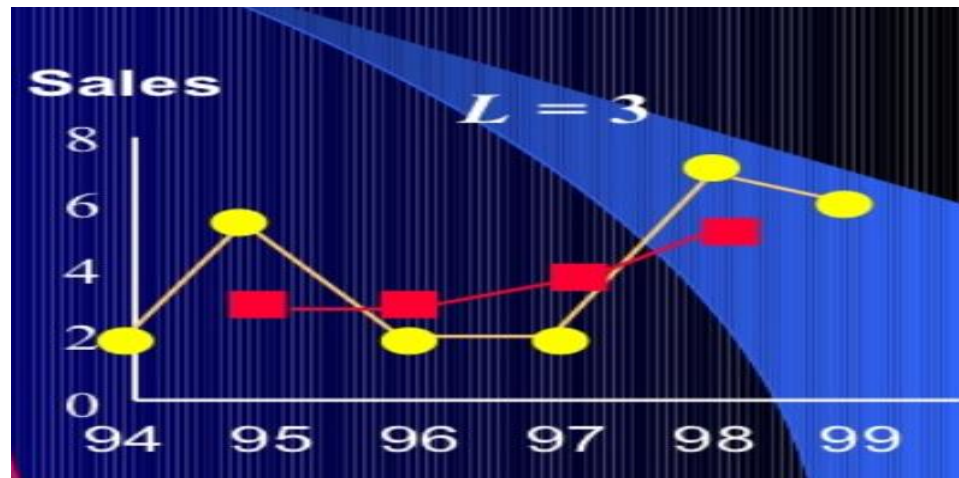


Moving Average

- ▶ Use neighborhood values to smooth out noise
- ▶ Typically used for time-series data
 - ▶ Use series of arithmetic means over time
 - ▶ Result depends on choice of length L for computing mean.
- ▶ Can also be used on spatial data such as images

Moving Average Example

Year	Sales	Moving Average
1994	2	NA
1995	5	3
1996	2	3
1997	2	3.67
1998	7	5
1999	6	NA



Using Z-scores to Detect Outliers

- ▶ Given an attribute X (e.g., Height) with a set of values, Z-score of value x_i is:

$$Z_i = \frac{x_i - \mu}{\sigma}$$

where μ is the mean of X , and σ is the standard deviation of X :

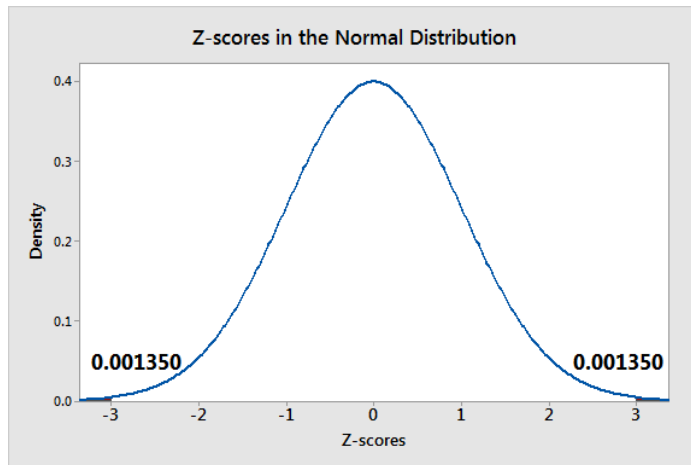
$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}} \text{ or } \sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N-1}}$$

- ▶ z_i measures how many σ 's away x_i is

Height	Z-score
1.5895	-0.34603
1.6508	-0.31975
1.7131	-0.29301
1.7136	-0.29283
1.7212	-0.28954
1.7296	-0.28595
1.7343	-0.28394
1.7663	-0.27020
1.8018	-0.25501
1.8394	-0.23888
1.8869	-0.21852
1.9357	-0.19757
1.9482	-0.19223
2.1038	-0.12551
10.8135	3.60910

Using Z-scores to Detect Outliers

- ▶ The further away x_i 's Z-score is from zero, the more unusual it is.
- ▶ A **standard cut-off value** for finding outliers is
 - ▶ Z-scores of **+/-3** or further from zero.
- ▶ Distribution of Z-scores in a standard normal distribution



Z-scores more extreme than +/-3 have a probability of 0.0027 (2×0.00135)

- ▶ Limitation:
 - ▶ If data don't follow the normal distribution, this approach might not be accurate.

Some Other Outlier Detection Methods

- ▶ The following methods consider multiple attributes together to determine whether an example is an outlier:
 - ▶ N = the number of examples
 - ▶ k = the number of attributes,

Outlier Detection Methods	Outlier Threshold
Mahalanobis Distance	$> \chi^2$ critical value with k degrees of freedom
Cook's Distance	$> 4/(N-k-1)$
Leverage	$> 2(k+1)/N$
Standardized DFFIT	$> 2\sqrt{(k+1)/N}$
Standardized DFBETA	$> 2/\sqrt{N}$

Data Generalization or Discretization

- ▶ Objective

- ▶ Smooth out local noise by generalization and discretization

- ▶ Methods for data generalization

- ▶ Nominal attributes

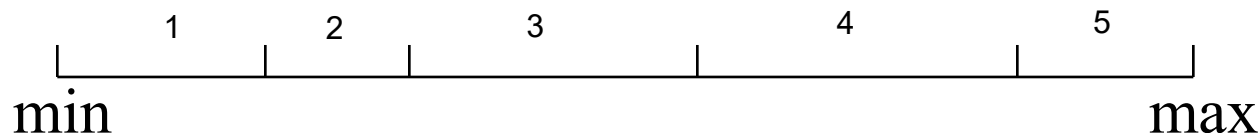
- ▶ Generalization

Toronto → Ontario → Central Canada → Canada

- ▶ Real-valued attributes

- ▶ Discretization:

- ▶ Partition the value range of an attribute into bins/intervals
 - ▶ label each bin with discrete values



Data Generalization or Discretization

- ▶ Other benefits
 - ▶ Reduce the number of distinct values of a feature so that the size of the search space for patterns is reduced.
 - ▶ Will see this in rule induction techniques
 - ▶ Model learned (e.g. decision tree) is smaller

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Data Reduction

- ▶ What is data reduction?

- ▶ Reduce the size of data while retaining its essential information

- ▶ Why data reduction?

- ▶ Efficiency

- ▶ A data set may be too large to fit in main memory.
 - ▶ The expected time for inducing a solution may be too long. Trade off accuracy for speed-up.

- ▶ Accuracy

- ▶ Better models may be found by using a reduced subset. E.g.,
 - ▶ Removing irrelevant redundant features mitigates the risk of overfitting
 - ▶ Too large data set may cause the model to fit too many exceptions if data are noisy

- ▶ Interpretability

- ▶ A larger dataset can often lead to larger (or deeper) decision trees
 - ▶ A smaller set of features makes the model more interpretable.

Data Reduction Operations

▶ Standard data form

<i>Case/Example</i>	<i>feature₁</i>	<i>...</i>	<i>feature_k</i>	<i>Class</i>
e_1	$V_{1,1}$	$\dots$	$V_{1,k}$	c_1
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
e_i	$V_{i,1}$	$\dots$	$V_{i,k}$	c_i
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
e_n	$V_{n,1}$	$\dots$	$V_{n,k}$	c_n

▶ Data reduction operations

- ▶ Feature reduction (reduce the number of columns)
- ▶ Case reduction (reduce the number of rows)

Types of Attributes (Features)

- ▶ Three types of attributes:
 - ▶ Nominal (symbolic, categorical) — values from an unordered set
 - ▶ Eg: {red, yellow, blue,}
 - ▶ Ordinal — values from an ordered set
 - ▶ Eg: {low, medium, high}
 - ▶ Continuous — real numbers
 - ▶ Eg: {-9.8, 3.9, 8.7, 19.1}
- ▶ Next: feature reduction for classification tasks.

Feature Reduction Techniques

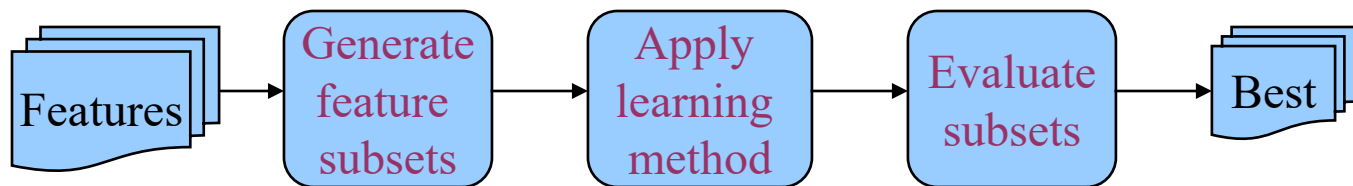
- ▶ Feature Selection
 - ▶ Selecting a subset of features with predictive performance comparable to the full set of features
- ▶ Feature Projection (also called Feature Extraction)
 - ▶ Mapping the original set of features into a new set of fewer features with new values.

Feature Selection

- ▶ Objective

- ▶ Find an optimal subset of features according to an objective function.

- ▶ An optimal subset selection



- ▶ A *practical objective* is to remove clearly extraneous features

- ▶ leaving the table reduced to manageable dimensions
 - ▶ not necessarily to select the optimal subset.

Feature Selection Methods

- ▶ Filter Methods: select a subset of original features.
 - ▶ Feature Selection from Means and Variances (✓)
 - ▶ Feature Selection by Mutual Information (✓)
 - ▶ Feature Selection by Chi-Square Test (✓)
 - ▶ Feature Selection by Decision Trees (✓)
 - ▶ Feature Selection by Rough Sets, etc.
- ▶ Wrapper Methods: feature selection is being “wrapper around” a learning algorithm.
 - ▶ This is the optimal method in the last slide.
 - ▶ Running time is long; infeasible in practice if there are many features.

Feature Selection from Means and Variances

▶ Principle

- ▶ Compute the means of a feature for each class, normalized by the variances;
- ▶ If the means are far apart, interest in a feature increases (the feature has potential in terms of distinguishing between classes);
- ▶ If the means are indistinguishable, interest wanes in that feature.

▶ Two intuitive methods

- ▶ Independent feature analysis ($\sqrt{}$)
 - ▶ Assuming the features are independent. Features are examined individually.
- ▶ Distance-based feature selection
 - ▶ Features are examined collectively.

▶ Limitation: only applied to continuous features.

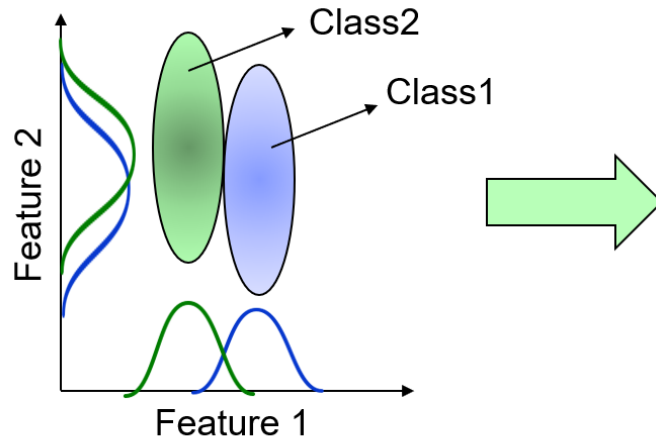
Independent Feature Analysis

- ▶ For a problem with two classes: C_1 and C_2 :
 - ▶ Compute $mean_1(f)$ and $mean_2(f)$: the means of feature f measured for C_1 and C_2
 - ▶ Compute $var_1(f)$ and $var_2(f)$: the variances of feature f measured for C_1 and C_2
 - ▶ Significance test (t-test):

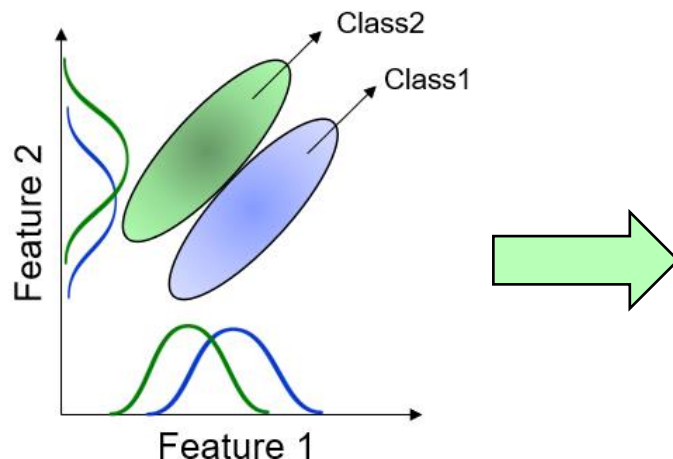
$$|mean_1(f) - mean_2(f)| > sig \times \sqrt{\frac{var_1(f)}{n_1} + \frac{var_2(f)}{n_2}}$$

- ▶ n_1 and n_2 are the numbers of cases in C_1 and C_2
 - ▶ $sig = 2$ for the 95% confidence level.
 - ▶ If the comparison fails the test, the feature can be deleted.
- ▶ For k classes, k pairwise comparisons are conducted for f .
 - ▶ Each pairwise comparison compares feature means for class C_i and $\neg C_i$ ($i=1, \dots, k$).
 - ▶ A feature is retained if it is significant for at least one of the pairwise comparisons.
- ▶ Limitations:
 - ▶ Only applies to numeric attributes
 - ▶ Treat each feature independently

Example (Independent Feature Analysis):



- Features are independent:
- Feature 1 provides better class separability.
- so Feature 2 is removed during selection.



- Features are not independent:
- They are collaboratively separable.
- so independent feature analysis cannot be used for feature reduction.

Feature Selection by Mutual Information

- ▶ Objective:
 - ▶ Select features according to the mutual information between a feature and the class variable.
- ▶ The mutual information (same as information gain) between the class variable y and a discrete feature x :

$$MI_x = \sum_j \sum_i [P(y = c_i, x = v_j) \times \log_2 \frac{P(y = c_i, x = v_j)}{P(y = c_i)P(x = v_j)}]$$

- ▶ $P(y=c_i)$ is the probability of examples in class c .
 - ▶ $P(x=v_j)$ is the probability that feature x takes on value v .
 - ▶ $P(y=c_i, x=v_j)$ is the probability of an example in class c and having v for feature x .
- ▶ MI measures the mutual dependence between the two variables x and y .
 - ▶ The higher the value, the more dependent one variable is on the other.

MI (example):

X (Feature)	Y (Prediction)
A	0
A	0
B	1
B	1
B	0

$$P(X = A) = \frac{2}{5}, \quad P(X = B) = \frac{3}{5}$$
$$P(Y = 0) = \frac{3}{5}, \quad P(Y = 1) = \frac{2}{5}$$

$$P(X = A, Y = 0) = \frac{2}{5}, \quad P(X = A, Y = 1) = 0, \quad P(X = B, Y = 0) = \frac{1}{5}, \quad P(X = B, Y = 1) = \frac{2}{5}$$

$$M_X = \sum_{x,y} P(x,y) \log_2 \frac{P(x,y)}{P(x)P(y)}$$

$$M_X = (0.4) \log_2 \frac{0.4}{0.4 \times 0.6} + (0.2) \log_2 \frac{0.2}{0.6 \times 0.4} + (0.4) \log_2 \frac{0.4}{0.6 \times 0.4}$$
$$M_X \approx 0.278 \text{ bits}$$

Feature Selection by Mutual Information

(Cont'd)

- ▶ *MI* is used to weight and rank features.
 - ▶ You can select the top k features with the highest weights.
- ▶ Suitable for nominal or discrete attributes.
 - ▶ For continuous features, a discretization algorithm can be applied first to convert a real-valued feature to a discrete-valued feature.
- ▶ Limitations:
 - ▶ Treat each feature independently.
 - ▶ No natural threshold for mutual information

Feature Selection Methods

- ▶ Filter Methods: select a subset of original features.
 - ▶ Feature Selection from Means and Variances (✓)
 - ▶ Feature Selection by Mutual Information (✓)
 - ▶ Feature Selection by Chi-Square Test (✓)
 - ▶ Feature Selection by Decision Trees (✓)
 - ▶ Feature Selection by Rough Sets, etc.
- ▶ Wrapper Methods: feature selection is being “wrapper around” a learning algorithm.
 - ▶ This is the optimal method in the last slide.
 - ▶ Running time is long; infeasible in practice if there are many features.

Feature Selection by Chi-Square Test

- ▶ Remove the features that are not co-related to the class attribute according to the χ^2 test
- ▶ The purpose of a χ^2 test is to determine whether two variables are co-related.
 - ▶ E.g., we want know if there is any relationship between the gender of undergraduate students in a university and their footwear preferences.
- ▶ Observations about the two variables in a sample are usually expressed in a contingency table:

	Sandals	Sneakers	Leather shoes	Boots	Other	Total
Male	6	17	13	9	5	50
Female	13	5	7	16	9	50
Total	19	22	20	25	14	100

Chi-Square Significance Test

- ▶ The **null hypothesis** is that *the two variables are unrelated* (that is, only randomly related).
- ▶ χ^2 test determines:
 - ▶ whether we should reject the null hypothesis and
 - ▶ *at what significance level (**p-value**)* we should reject the null hypothesis.
 - ▶ The **p-value** is the probability of obtaining the observed sample results (or a more extreme result) when the null hypothesis is actually true
 - ▶ Usually, p-value should be **at most 0.05** in order to reject the null hypothesis.

Chi-Square Significance Test

Footwear preference

	Sandals	Sneakers	Leather shoes	Boots	Other	Total
Male	6	17	13	9	5	50
Female	13	5	7	16	9	50
Total	19	22	20	25	14	100

- ▶ For the example above,
 - ▶ The null hypothesis is that *Gender* is unrelated with *Footwear preference*
 - ▶ But the χ^2 test shows that we should reject this hypothesis at the significance level of 0.01(**p-value**)

How to Calculate χ^2

- ▶ Given the contingency table:

	Sandals	Sneakers	Leather shoes	Boots	Other	Total
Male	6	17	13	9	5	50
Female	13	5	7	16	9	50
Total	19	22	20	25	14	100

- ▶ Compute the expected frequency for each cell
assuming null hypothesis

- ▶ The expected frequency of cell_{i,j} is

$$E_{ij} = \frac{\text{the total of row } i \times \text{the total of column } j}{\text{sample size}}$$

- ▶ For example, the expected frequency of the upper left cell is $\frac{50 \times 19}{100}$

How to Calculate χ^2 (Cont'd)

- ▶ Compute the chi-square value for the table

	Sandals	Sneakers	Leather shoes	Boots	Other	Total
Male	6	17	13	9	5	50
Female	13	5	7	16	9	50
Total	19	22	20	25	14	100

- ▶ Let O_{ij} denote the observed value in cell _{i,j}

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

- ▶ For example, the chi-square value of the above table is 14.026

How to Calculate χ^2 (Cont'd)

- ▶ Calculate the degrees of freedom for the table

	Sandals	Sneakers	Leather shoes	Boots	Other	Total
Male	6	17	13	9	5	50
Female	13	5	7	16	9	50
Total	19	22	20	25	14	100

$$df = (r-1)(c-1)$$

- ▶ where r is the number of rows and c is the number of columns (without the row or column for totals)
- ▶ For example, the degrees of freedom for the above table is 4.
- ▶ This is because, given row or column totals, all but one of the values in a given row or column are free to vary.

How to Calculate χ^2 (Cont'd)

- ▶ Using the chi-square table to determine the *p-value* for rejecting the null hypothesis

df	P = 0.05	P = 0.01	P = 0.001
1	3.84	6.64	10.83
2	5.99	9.21	13.82
3	7.82	11.35	16.27
4	9.49	13.28	18.47
5	11.07	15.09	20.52
...	...	...	...

- ▶ The table lists the critical values (i.e., thresholds)
- ▶ The calculated chi-square value for a contingency table must be greater than the critical value corresponding to the *df* of the table and a *p-value* (e.g., 0.05) in order to reject the null hypothesis at the significance level (p-value).

Feature Selection Methods

- ▶ Filter Methods: select a subset of original features.
 - ▶ Feature Selection from Means and Variances (✓)
 - ▶ Feature Selection by Mutual Information (✓)
 - ▶ Feature Selection by Chi-Square Test (✓)
 - ▶ Feature Selection by Decision Trees (✓)
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 - ▶ This is the optimal method in the last slide.
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Feature Selection by Decision Trees

▶ Objective

- ▶ Decision tree learning methods integrates feature selection to their algorithms and decision tree is a fast learning method.
- ▶ Make use of the decision tree learning technique to select features from a data set for other learning methods, such as neural networks, that take substantially more time to search their solution space.
 - ▶ Decision tree learning is a relatively fast learning method.

▶ Method

- ▶ Apply a decision tree learning algorithm to the data set to generate a decision tree.
- ▶ Select features that appear in the tree.

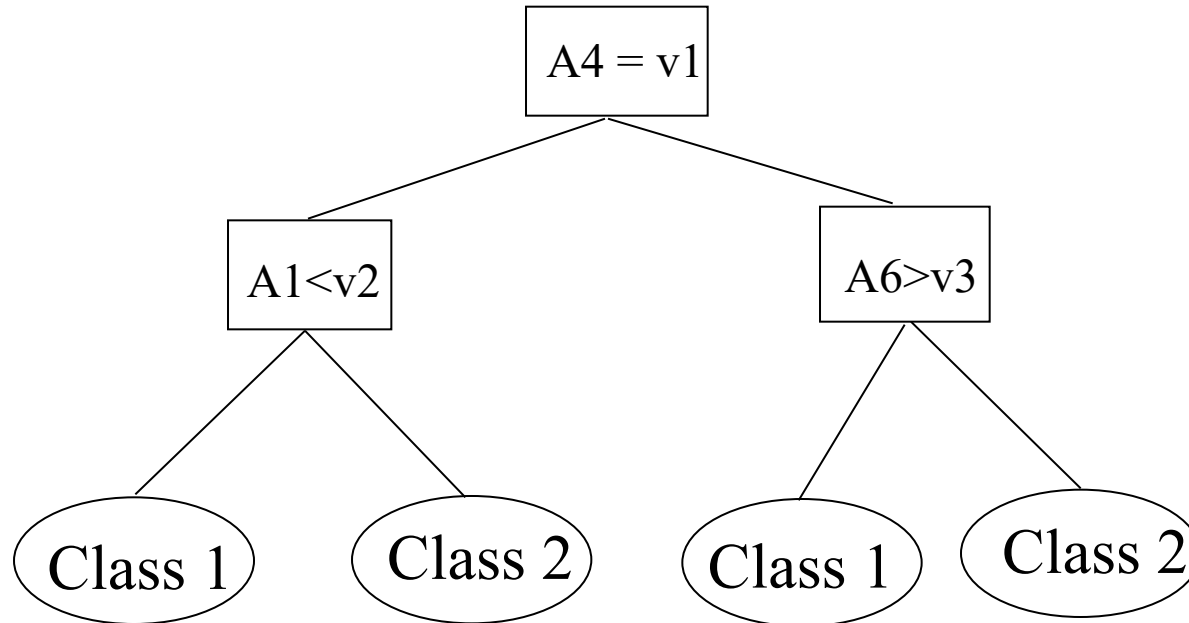
▶ Advantage

- ▶ *Context sensitive*. Tree methods evaluate candidate features in the context of related features that have already been selected.

Example of Feature Selection Using Decision Tree

Initial attribute set:

$\{A1, A2, A3, A4, A5, A6\}$



-----> Reduced attribute set: $\{A1, A4, A6\}$

Feature Reduction Techniques

- ▶ Feature Selection
 - ▶ Selecting a subset of features with predictive performance comparable to the full set of features
- ▶ Feature Projection (also called Feature Extraction)
 - ▶ Mapping the original set of features into a new set of fewer features with new values.

Feature Projection

▶ Objective

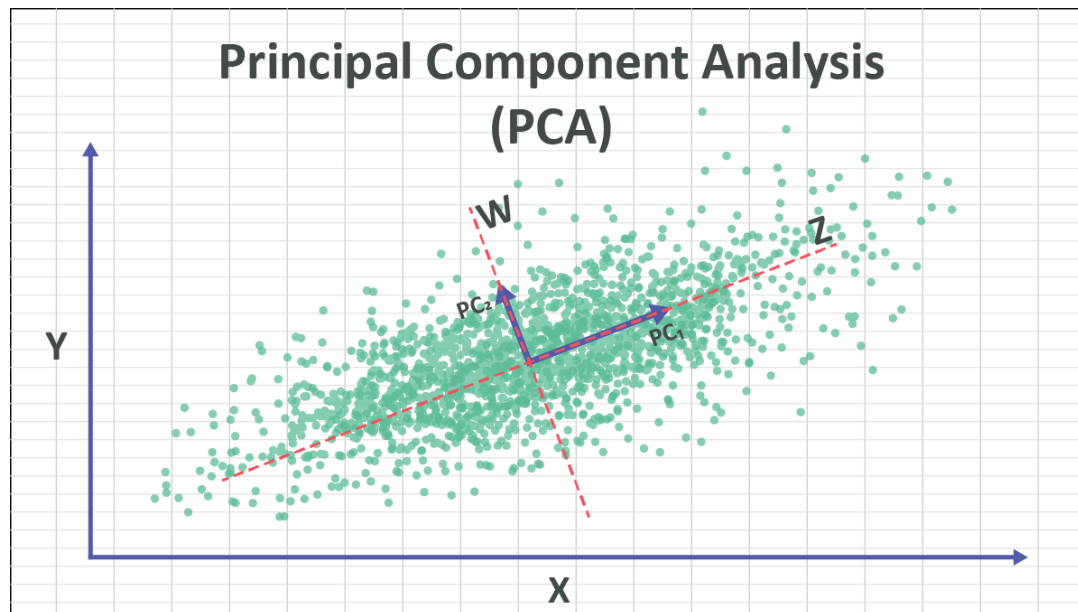
- ▶ Map the original features into a smaller set of newly created features which hopefully retains most of the relevant information from *all* the original features.

▶ Methods

- ▶ Principal component analysis (PCA)
- ▶ Neural autoencoder
- ▶ t-distributed stochastic neighbor embedding (t-SNE)
- ▶

Feature Projection

- ▶ Principal Component Analysis:
 - ▶ PCA transforms correlated features into a set of uncorrelated components.
 - ▶ It captures the directions (principal components) with the highest variance in the data.
 - ▶ It replaces original features with top k principal components that explain the most variance.



Feature Projection

- ▶ Neural Autoencoder:
 - ▶ It learns to compress input data into a smaller representation and then reconstruct it.
 - ▶ The goal is to capture the most important information in a low-dimensional form.
 - ▶ PCA is linear, while autoencoders learn nonlinear feature projections.

